

ASSESSMENT TOOL 1

COURSE CODE: CSA14 COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO1: Apply lexical analysis for token specification and recognition using LEX tool.

(BL3) Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:50 Mode: In Class Hours : 60 mts Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

*I **J.Mohamed Aseem(192511258)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

J.Mohamed Aseem

Signature of the student:

Annexure A

ANALYTICAL PROBLEM SOLVING

1	Trace the compilation of the arithmetic expression $R = (x + y) * (y - z) + 5$ through the major phases of a compiler. a) Identify and classify all tokens produced during lexical analysis. b) Construct the corresponding syntax tree during the parsing phase. c) Generate the three-address intermediate code for the expression. d) Apply suitable code optimization techniques and show the optimized intermediate representation. e) Generate a possible target-code sequence and explain how the final code contributes to efficient execution.

2	For each of the following Regular Expressions, construct the language. Write any FIVE valid strings for each Regular Expression. a) $(00 + 0000^*)^*$ b) $(00^* + 00^*)^* 0000000$ c) $(00^*000^*)^* 00000000 (00^* + 00^*)^*$ d) $(00 + 00000)^*$ e) $(0000000)^* + (00^*000^*)^*$
3	i) Find the number of tokens in the following C statement is <pre>main() { int a=10; char b = "abc"; in t c = 30; ch ar d = "xyz"; in /* comment */ t m = 40.5; }</pre> ii) Identify the lexical errors in the following C statement is <pre>void main() { int x=10, y=20; char * a; a= &x; x= 1xab; }</pre>
4	Find a regular expression corresponding to each of the following subsets of $\{0, 1\}^*$: i) The language of all strings that have an even number of 0's. ii) The language of all strings that contain at least one 1. iii) The language of all strings in which both the number of 0's and the number of 1's are odd.
	iv) The language of all strings that start with 1 and end with 0. v) The language of all strings that do not contain consecutive 1's.

5

Consider a lexical analyzer for a simplified programming language. The language consists of the following tokens:

1. Keywords: if, else, while, return, int, float
2. Operators: +, -, *, /, =, ==, <, >
3. Separators: (,), {, }, ;, ,
4. Identifiers: A sequence of letters followed by a sequence of digits, e.g., var123
5. Integer Literals: A sequence of digits, e.g., 12345
6. Floating-point Literals: A sequence of digits followed by a dot and another sequence of digits, e.g., 123.45

The lexical analyzer needs to tokenize the following input string: int var123 = 12345; float var456 = 123.45; if (var123 == var456) { return; } **Questions:**

- a) How many tokens are there in the given input string?
- b) Provide a list of tokens categorized by their types (keywords, operators, separators, identifiers, integer literals, floating-point literals).
- c) Assuming a finite state machine (FSM) is used for lexical analysis, estimate the number of state transitions required to tokenize the given input string. Provide your reasoning based on the transitions needed for each token type.

ANSWERS

1) trace the compilation of $R = (x+y) * (y-2) \div 5$

a) Lexical Analysis - Tokens:-

Lexeme

Token

R

Identifier

=

assignment operator

(

Left Parenthesis

x

Identifier

+

Arithmetic operator

y

Identifier

)

Right Parenthesis

*

Multiple operator

(

Left Parenthesis

+

Arithmetic operator

5

integer

Total token Sequence:-

Identifier - 3

operators - 3

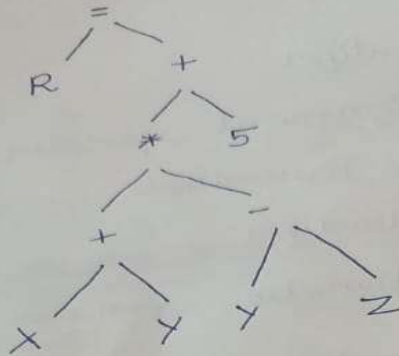
Parenthesis - 4

Integer - 1

total = 11

b) Syntax tree:-

$$R = ((X + Y) * (Y - Z)) + 5$$



c) Three address code:-

$$T_1 = X + Y$$

$$T_2 = Y - Z$$

$$T_3 = T_1 * T_2$$

$$T_4 = T_3 + 5$$

$$R = T_4$$

d) code Optimization:-

$$T_1 = X + Y$$

$$T_2 = Y - Z$$

$$T_3 = T_1 * T_2$$

$$R = T_3 + 5$$

4) Possible target code:-

LOAD R1, X

ADD R1, Y

LOAD R2, Y

SUB R2, Z

MUL R1, R2

ADD R1, 5

STORE R, R1

2) Regular expression:-

a) $(\diamond + \diamond\diamond\dots)^*$

b) $(\diamond^* + \diamond^*)^* \dots$

c) $(\diamond^* \diamond^*)^* \dots$

d) $(\diamond + \diamond\diamond\dots)^*$

e) $(\diamond\diamond\diamond\diamond)^* + \dots$

3) Lexical Analysis:-

i.) Find the number of tokens:-

Main C)

{

int a=10;

char b="abc"

char d="xyz"

in /*comment*

5)



3.) S-

A-

B-

A-

F

F

token count

tokens

```

Main C)
{
int a=10;
char b="abc"
int c=10;
char d="xyz"
in /*comment*/ t m=105;
}
total
    
```

3
 1
 5
 5
 6
 6
 1
 1
 33

ii) Identify the lexical error:-

```

void Main C)
{
int x=10, y=20;
char *a;
a = &x;
x = 1xab;
}
    
```

error:-

x = 1xab

char *a;

a = &x;

Lexical error:-

1xab → Invalid.

4) Regular expression for Given Language:

i) String having an even Number of 0's

$1^*(01^*01^*)^*$

ii) String containing at least one 1 :-

$0^*1(0|1)^*$

iii) both Number of 0's and Number of 1's are even

$(00|11)^*01(00|11)^*$

iv) String starting with 1 & ending with 0 :-

$1(0|1)^*0$

v) String that Do not contain consecutive

$(01|10)^*(\epsilon|1)$

5) Number of token :-

a) i) `int var123 = 12345;`

tokens :-

- 1) int
- 2) var123
- 3) =
- 4) 12345
- 5) ;

ii)

`float var456 = 123.45;`

- 6) float
- 7) var456
- 8) =
- 9) 123.45
- 10) ;

iii)

`if (var123 == var456) { return`

- 11) if
- 12) (
- 13) var123
- 14) ==
- 15) var456
- 16))
- 17) {
- 18) return
- 19) ;
- 20) }

No. of. tok

5.)
b) Categorization of token:-

1.) Keywords:-

int
float
if
return

No. of keyword = 4
2.) operator:-

=
=

No. of operator = 3
3.) Separators:-

;
;
{
}
;
}

No. of separator = 7

4.) Identifiers:-

Var123

Var456

Var123

Var456

No. of token = 4

5.) Integer Literals:-

12345

Number = 1

6.) Floating Point Literals:-

123.45

No = 1

5.) FSM state transition:-

int $\rightarrow$ 3

float $\rightarrow$ 5

if $\rightarrow$ 2

return $\rightarrow$ 6

$$tot = 3 + 5 + 2 + 6 = 16$$

Var 123 $\rightarrow$ 6

Var 456 $\rightarrow$ 6

Var 123 $\rightarrow$ 6

Var 456 $\rightarrow$ 6

$$6 \times 4 = 24$$

= $\rightarrow$ 1

= $\rightarrow$ 1

= $\rightarrow$ 2

$$1 + 1 + 2 = 4$$

$\therefore$ () $\leq$;

7 transition

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation